TEXORA · LECTURE SHEET

Yarn Flow Charts, Mixing and Bale Management

How raw bales become yarn: the machine sequences for carded and combed cotton, the long-staple routes in outline, and the mixing and bale management on which every later process depends.

TE-2101 · SPINNING TECHNOLOGY-I · TOPIC 1 · LEVEL 2 TERM 1

1. Why a flow chart matters

Spinning is a **sequence**, not a collection of machines. Each machine can only correct part of what the previous machine left undone, and none can fully repair an earlier fault. A **flow chart** shows, in order, every operation from bale to packaged yarn: where waste is removed, where doubling and drafting occur, and what the yarn can achieve. Three ideas run through every chart: **mass reduces progressively** by drafting (lap → sliver → roving → yarn); **cleaning is front-loaded** (most trash leaves in the blow room and at the card); and **evenness is built by doubling**, several strands combined so thick and thin places average out, above all at the draw frame. Short- and medium-staple cotton for ordinary counts follows the **carded route**; the same cotton for finer, stronger yarn follows the **combed route**; long-staple fibres need length-adapted routes (Section 4).

2. The carded cotton route, step by step

1. **Bale opening and mixing.** Bales selected for one mixing are opened and laid down together (Sections 5–7). The fibre is still in compressed tufts with trash, leaf, seed fragments and dust.
2. **Blow room.** Opening and cleaning machines beat and air-wash the tufts progressively smaller, remove typically 4–8 % waste, blend the bales, and deliver a rolled **lap** of set weight per metre — or, in chute-feed lines, an even mat direct to the card.
3. **Carding.** The card opens fibre to *individual fibres* between wire-covered cylinder, flats and doffer, removes most remaining trash and neps and some very short fibre, begins parallelisation, and delivers **card sliver** in a can. It is the last effective cleaning point.
4. **Draw frame (two passages).** In each passage 6–8 slivers are doubled and roller-drafted (draft ≈ doublings, so weight per metre is roughly kept). Drafting straightens hooked fibres — the first passage is fed to act on the trailing hooks left by the card — and doubling averages irregularity. Output: even, parallel **drawn sliver**.

5. **Simplex (roving frame).** Drawn sliver is drafted (roughly 5–15) to a thin **roving**, given slight protective twist and wound on bobbins, fine enough for the ring frame's final draft yet strong enough to unwind without stretching.
6. **Ring frame.** Roving is drafted to the required count (draft roughly 10–50), twist is inserted by ring and traveller to give strength, and yarn is wound on cops. Count, twist and strength are finally fixed here.
7. **Winding.** Cops hold only a short length, so winding clears faults with electronic clearers, splices lengths together, waxes knitting yarn if required, and builds large cones or cheeses for weaving, knitting or dispatch.

3. The combed cotton route

For fine counts, quality shirtings, sewing threads and high-grade knitwear the carded route is not clean or parallel enough. The combed route is identical except for an inserted block between carding and the simplex:

1. **Bale → blow room → card** — as above, the card worked gently for a clean, even sliver.
2. **Pre-comber drawing.** One draw-frame passage straightens fibres and reverses the hook direction so the comber can act on the majority of hooks.
3. **Lap former.** About 20–28 drawn slivers are doubled, lightly drafted and combined into an even **comber lap** of comber-head width (commonly about 300 mm, at 60–80 g/m). Any thick place would overload the comber needles, so evenness here is critical.
4. **Comber.** A needle cylinder and top comb comb the lap fringe; short fibres, neps and trash are rejected as **noil**, typically 10–25 % of the feed, and long fibres are delivered straight and parallel as combed sliver (full treatment in TE-2201).
5. **Post-comber drawing.** One or two passages remove the periodic irregularity left by the comber's piecing-up and restore evenness.
6. **Simplex → ring frame → winding** — as in the carded route, but finer counts are now attainable because short fibre has been removed.

The cost is extra machines plus the deliberate noil loss — justified only when yarn price rewards the strength, evenness, lustre and cleanliness gained: the mill *buys quality with raw material*.

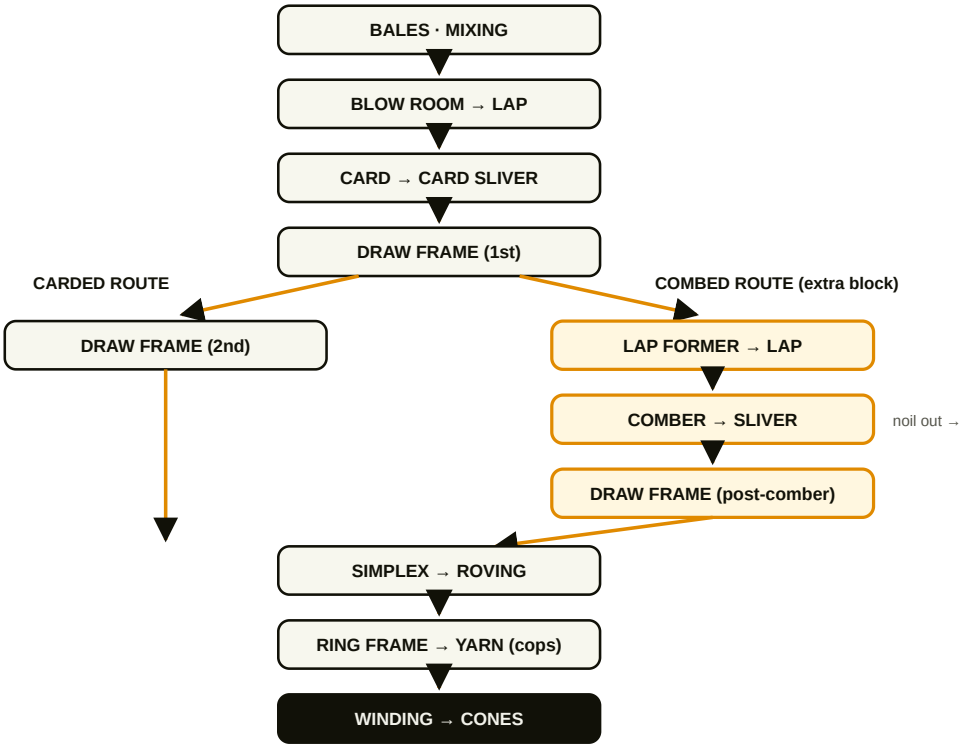


Figure 1 — Cotton flow chart. Both routes share bale, blow room, card and first drawing. The carded route goes to a second drawing and the simplex; the combed route (amber) inserts lap former and comber, rejects noil, and draws again before the simplex.

COMPARISON	CARDED YARN	COMBED YARN
After the card	2nd drawing → simplex	Pre-comber drawing → lap former → comber → post-comber drawing(s) → simplex
Short fibre	Retained	Removed as noil, 10–25 %
Typical counts	Roughly Ne 6–40	Roughly Ne 30–100+
Character / uses	Adequate strength, hairier; denim, ordinary knitwear	Stronger, evener, smoother; fine shirting, sewing thread

4. Long-staple routes in brief

Jute: batching and selection → **softener / spreader** (oil-and-water emulsion; piled 24–48 h to soften the stiff ribbons) → **breaker card** → **finisher card** → **drawing frames** (2–3 passages) → **spinning frame** → winding. No roving stage: drawing delivers sliver the spinning frame drafts directly. **Worsted** (long wool): scouring → carding → repeated **gilling** → combing (noil removed) → gilling and drawing to roving → spinning → winding; yarn smooth and lean. **Woollen** (short wool): scouring → blending and oiling → **condenser card**, which itself divides the web into tapes and rubs them into rovings → spinning → winding. No combing; short fibres kept; yarn bulky and warm.

5. Mixing — objects and methods

A single bale is never spun alone, and one growth is rarely uniform enough to spin alone. **Mixing** combines bales or lots (and, in its intimate form, **blending** combines different fibre types, e.g. cotton–polyester). Its objects:

- **Uniformity** — bale-to-bale variation in length, fineness and strength averages out, so count, strength and dye uptake stay constant week to week.
- **Process stability** — a well-mixed lot runs with fewer ends down at card, draw frame and ring frame.
- **Economy** — expensive long-staple cotton is extended with cheaper cotton while still meeting specification, if proportions are controlled (worked examples below).
- **Required properties** — blends give what no single fibre gives: strength, easy care and comfort together; a correcting proportion for maturity or colour.
- **Shade control** — colour and maturity differences that would show as barré (stripiness) after dyeing are diluted before spinning.

The governing principle: **as many bales, as intimately mixed, as early as possible** — bale-stage variation cannot be removed later, only diluted, and dilution is cheapest at the start. **Manual (stack) mixing:** tufts are stacked in horizontal layers, one from each stock in turn, then fed by cutting *vertically* through all layers so every slice contains every stock — thorough if disciplined, but slow, dusty and labour-heavy. **Machine / automatic mixing:** the plucker takes a little from every bale each pass, blending hoppers dose stocks by weight or timed feed, and every later doubling mixes further. Proportions are held automatically over laydowns of 20–60+ bales — the production standard.

6. Bale management

Bale management is the system — buying, testing, storing, selecting, issuing — that keeps the mixing constant day after day although the underlying bales keep changing.

6.1 Selection by fibre properties

Bales are classed by instrument testing (typically HVI) and grouped so each laydown's averages match the yarn target:

- **Staple length and uniformity** — chief determinant of spinnable count and strength; widely different lengths draft differently and must not be mixed carelessly.
- **Fineness / micronaire** — an airflow value combining fineness and maturity; it sets the number of fibres in the yarn cross-section, hence evenness, strength and dye uptake.

- **Strength** — weak bales cap yarn strength whatever the spinning; the mixing average must clear the specification with margin.
- **Maturity** — immature fibres form neps, break in processing and dye light (white specks); immature bales are limited or excluded from quality mixings.
- **Trash and colour / grade** — trashy bales overload blow room and card; colour must be balanced across the laydown to avoid shade variation.

The store holds a bale inventory sorted by these properties and the mixing is *formulated* like a recipe to hit target averages at acceptable cost. When a stock runs out it is replaced **gradually** — only a small fraction of the laydown at a time — so yarn character never jumps (the transition rule).

6.2 Laydown and conditioning

The **laydown** is the physical arrangement of the chosen bales for one mixing: one or two rows with sides exposed to the plucker. Rules: use as many bales as the system allows; take every group in the formulation in proportion; **interleave** each stock through the laydown rather than blocking it, so every plucker pass samples all stocks; keep height and tuft size uniform so feed does not surge (Figure 2). Before opening, bales are **conditioned** — rested at least 24 hours in the processing atmosphere (about 20–25 °C, 60–65 % RH) with coverings loosened, to reach moisture equilibrium (cotton regain about 7–8.5 %) and relax from compression. Skipping conditioning gives variable opening, fly, static and uneven laps that are easily mistaken for machine faults.

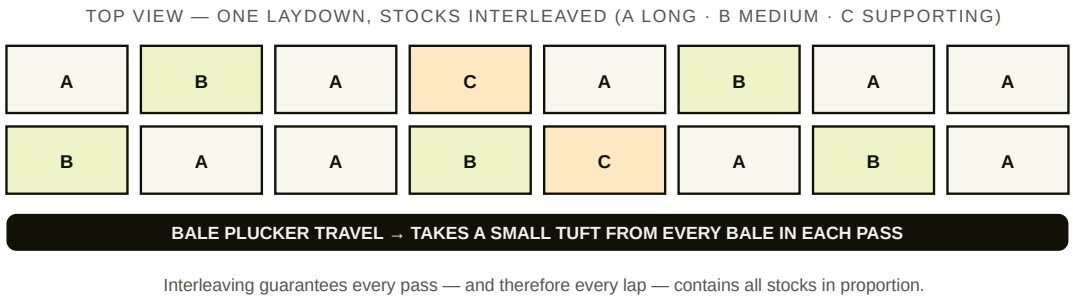


Figure 2 — Bale laydown plan (top view). Stocks are interleaved, not blocked together, so automatic plucking samples the full mixing continuously.

7. Mixing quality and evenness

Mixing quality is judged downstream — lap weight, sliver count, yarn count CV %, strength and dyed shade, within and between lots. Its enemies: too few bales, abrupt wholesale stock change, blocked laydowns, unequal tuft size, unrecorded substitution. Drafting can only redistribute fibres already present in roughly the right proportions: the evenness ceiling of the yarn is set at the laydown and the card.

WORKED EXAMPLE — BALES IN A 40-BALE MIXING

A mixing is formulated as 60 % stock A, 25 % stock B, 15 % stock C, in a 40-bale laydown. Bales per stock:

A: 40 × 60 ÷ 100 = 40 × 0.60 = **24 bales**. B: 40 × 0.25 = **10 bales**. C: 40 × 0.15 = **6 bales**.

Check: 24 + 10 + 6 = 40 bales. ✓ Interleave the three stocks through the laydown (Figure 2), do not block them.

WORKED EXAMPLE — WEIGHTED MIXING COST

Stocks cost A = 520, B = 460, C = 400 Tk/kg, mixed 60 / 25 / 15 %. Mixing cost is the proportion-weighted average:

$$\text{Mixing cost} = 0.60 \times 520 + 0.25 \times 460 + 0.15 \times 400$$

0.60 × 520 = 312; 0.25 × 460 = 115; 0.15 × 400 = 60. Total = 312 + 115 + 60 = **487 Tk/kg** — 33 Tk/kg below stock A alone, while 60 % premium stock still dominates spinning behaviour.

Glossary

Flow chart — the ordered machine sequence from bale to yarn. **Carded / combed yarn** — yarn spun without combing / after combing has removed short fibre. **Noil** — short-fibre comber waste. **Lap** — the even fibre sheet from the blow room (or fed to the comber). **Sliver** — the untwisted strand from card, draw frame and comber. **Roving** — the lightly twisted simplex strand, ready for ring spinning. **Mixing / blending** — combining bales or lots / intimately combining different fibres in set proportions. **Laydown** — the bales arranged for one mixing. **Micronaire** — airflow value for fineness combined with maturity. **Conditioning** — resting bales to moisture equilibrium before opening. **Barré** — stripe-like dyed shade variation from uneven mixing or maturity.

Key points

• Carded: bale → blow room → card → draw frame ×2 → simplex → ring frame → winding. Combed inserts lap former + comber + extra drawing between card and simplex.

• Combing removes 10–25 % as noil and buys strength, evenness and fineness at the cost of yield.

• Jute: softening, breaker and finisher cards, drawing, spinning — no roving stage. Worsted: gilling + combing. Woollen: condenser card, short fibres kept.

• Mixing averages bale variation, stabilises processing, controls cost and shade — as many bales, as intimately, as early as possible.

• Bale management = HVI selection (length, micronaire, strength, maturity, trash) + interleaved

laydown + 24 h conditioning; replace stocks gradually.

• Bales per stock = total bales × proportion; mixing cost = Σ (proportion × price).

Practice

1. Draw the combed cotton flow chart and mark every point where waste is removed, naming each waste.
2. Stocks priced 540, 480 and 420 Tk/kg are mixed 50 % / 30 % / 20 %. Calculate the mixing cost per kilogram, showing each term.
3. A mill changes its entire laydown overnight and the next day's yarn dyes streaky. Explain what went wrong and the correct procedure.
4. Why has the classic jute route no simplex-equivalent stage, and which woollen machine does the work of both draw frame and simplex?
5. State the five objects of mixing and, for two of them, the fault that appears if the object is neglected.
6. A 48-bale laydown is planned at 50 % / 30 % / 20 %. Show why bale counts alone cannot hold these proportions, and give a practical solution.

Answer key

1. Bales/mixing → blow room (blow-room waste: trash, dust) → card (flat strips and card waste: short fibre, trash, neps) → drawing → lap former → comber (noil: short fibre, neps, trash) → drawing → simplex → ring frame → winding (clearer cuts: faulty yarn lengths).
2. 0.50 × 540 = 270; 0.30 × 480 = 144; 0.20 × 420 = 84. Total = 270 + 144 + 84 = **498 Tk/kg**.
3. The wholesale change broke the transition rule: length, micronaire, maturity and colour jumped together, changing drafting behaviour and dye uptake — seen as barré. Correct procedure: HVI-test the shipment, reformulate to hold target averages, replace only a small fraction of the laydown at a time.
4. Jute drawing delivers a coarse sliver the spinning frame drafts directly to yarn count, so no roving package is needed. In the woollen system the **condenser card** divides its web into tapes and rubs them into rovings.
5. Uniformity; process stability; economy; required properties; shade control. E.g. neglected uniformity → count/strength CV and ends down rise; neglected shade control → barré after dyeing.
6. 50 % of 48 = 24; 30 % of 48 = 14.4; 20 % of 48 = 9.6 — bales are whole, so counts alone cannot be exact. Solutions: use a divisible laydown size (50 bales → exactly 25 / 15 / 10), or hold proportions by plucker passes / hopper weighting, not bale count.